

Télémétrie des équipements réseau

2 Installation de l'environnement du TP

2.1 Installation de pyenv et de Python 3.11

```
test@0-179:~/gnmic$ rm -rf /home/test/.pyenv
test@0-179:~/gnmic$ curl -fsSL https://pyenv.run | bash
Clonage dans '/home/test/.pyenv'...
remote: Enumerating objects: 1386, done.
remote: Counting objects: 100% (1386/1386), done.
remote: Compressing objects: 100% (716/716), done.
```

Modifiez votre .bashrc pour activer pyenv au démarrage de votre shell (à mettre avant fzf):

```
GNU nano 7.2 /home/test/.bashrc
# ~/.bashrc: executed by bash(1) for non-login shells.
# see /usr/share/doc/bash/examples/startup-files (in the package bash-doc)
# for examples
export PYENV_ROOT="$HOME/.pyenv"
[[ -d $PYENV_ROOT/bin ]] && export PATH="$PYENV_ROOT/bin:$PATH"
eval "$(pyenv init - bash)"
eval "$(pyenv virtualenv-init -)"

# If not running interactively, don't do anything
case $- in
  *i*) ;;
  *) return;;
esac

# don't put duplicate lines or lines starting with space in the history
```

installez la version de Python 3.11 (gnmic ne fonctionne pas à la date de création du TP avec Python3.12):

Tout d'abord, installez les dépendances nécessaires à la compilation de Python 3.11 :

```
test@0-179:~/gnmic$ nano ~/.bashrc
test@0-179:~/gnmic$ sudo apt install -y make build-essential libssl-dev zlib1g-dev \
libbz2-dev libreadline-dev libsqlite3-dev wget curl llvm \
libncurses5-dev libncursesw5-dev xz-utils tk-dev libffi-dev \
liblzma-dev python3-openssl libssh-dev openswitch-switch sshpass
[sudo] Mot de passe de test :
Lecture des listes de paquets... Fait
Construction de l'arbre des dépendances... Fait
Lecture des informations d'état... Fait
```

Répérez la dernière version de Python 3.11 et installez-la :

```
test@0-179:~/gnmic$ pyenv install -l | grep 3.11
pyenv install 3.11.11
pyenv virtualenv 3.11 netlab
pyenv activate netlab
3.11.0
3.11-dev
3.11.1
3.11.2
3.11.3
```

2.2 Installation de Netlab

Installer NetLab en suivant les instructions suivantes

```
Installed Python-3.11.11 to /home/test/.pyenv/versions/3.11.11
(netlab) test@0-179:~/gnmic$ pip install --upgrade pip
pip install ansible-pylibssh networklab[ALL] ansible
Requirement already satisfied: pip in /home/test/.pyenv/versions/3.11.11/
site-packages (24.0)
Collecting pip
  Using cached pip-25.1.1-py3-none-any.whl.metadata (3.6 kB)
Using cached pip-25.1.1-py3-none-any.whl (1.8 MB)
Installing collected packages: pip
```

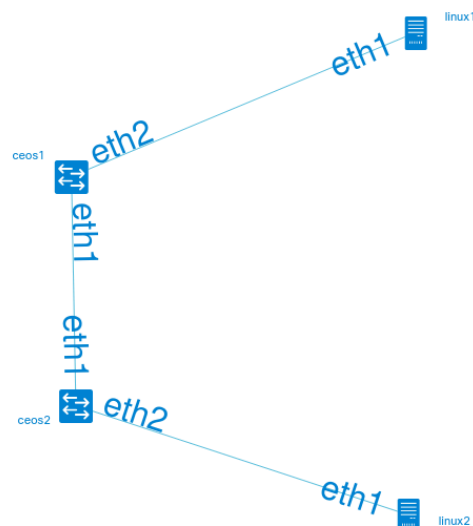
2.3 Installation de gnmic

Créez une directory "tpgnmic", placez-vous dans cette directory et installez gnmic :

```
(netlab) test@0-179:~/gnmic$ mkdir tpgnmic
cd tpgnmic
bash -c "$(curl -sL https://get-gnmic.openconfig.net)"
Downloading https://github.com/openconfig/gnmic/releases/download/v0.41.0/gnmic_0.41.0_linux_x86_64.
tar.gz
Preparing to install gnmic 0.41.0 into /usr/local/bin
gnmic installed into /usr/local/bin/gnmic
version : 0.41.0
commit : 932a2403
date : 2025-04-15T15:17:24Z
gitURL : https://github.com/openconfig/gnmic
docs : https://gnmic.openconfig.net
(netlab) test@0-179:~/gnmic/tpgnmic$
```

3 Création d'une infrastructure containerisée de routeurs

On va générer un fichier topology.yml pour créer une infrastructure de test avec 2 routeurs Arista et 2 clients Linux



```
(netlab) test@0-179:~/gnmic/tpgnmic$ netlab report addressing
Node/Interface      IPv4 address      IPv6 address Description
=====
ceos1 (10.0.0.1/32)
  Ethernet1          10.1.0.1/30       ceos1 -> ceos2
  Ethernet2          172.16.0.1/24     ceos1 -> linux1
ceos2 (10.0.0.2/32)
  Ethernet1          10.1.0.2/30       ceos2 -> ceos1
  Ethernet2          172.16.1.2/24     ceos2 -> linux2
linux1
  eth1               172.16.0.3/24     linux1 -> ceos1
linux2
  eth1               172.16.1.4/24     linux2 -> ceos2
```

```
provider clab: executing sudo -E containerlab deploy --reconfigure -t clab.yml
14:48:32 INFO Containerlab started version=0.68.0
14:48:32 INFO Parsing & checking topology file=clab.yml
14:48:32 INFO Removing directory path=/home/test/gnmic/tpgnmic/clab-tpgnmic
14:48:32 INFO Creating docker network name=netlab_mgmt IPv4 subnet=192.168.121.0/24 IPv6 subnet="" MTU=0
14:48:32 INFO Pulling registry.iutbeziers.fr/network-multitool:mai2025 Docker image
14:48:35 INFO Done pulling registry.iutbeziers.fr/network-multitool:mai2025
14:48:35 INFO Creating lab directory path=/home/test/gnmic/tpgnmic/clab-tpgnmic
14:48:35 INFO Creating container name=linux1
14:48:35 INFO Creating container name=linux2
14:48:35 INFO Creating container name=ceos2
14:48:35 INFO Creating container name=ceos1
14:48:36 INFO Running postdeploy actions for Arista cEOS 'ceos2' node
14:48:36 INFO Created link: ceos1:et1 ----- ceos2:et1
14:48:36 INFO Created link: ceos1:et2 ----- linux1:eth1
14:48:36 INFO Running postdeploy actions for Arista cEOS 'ceos1' node
14:48:36 INFO Created link: ceos2:et2 ----- linux2:eth1
14:49:03 INFO Adding host entries path=/etc/hosts
14:49:03 INFO Adding SSH config for nodes path=/etc/ssh/ssh_config.d/clab-tpgnmic.conf
```

Name	Kind/Image	State	IPv4/6 Address
clab-tpgnmic-ceos1	ceos registry.iutbeziers.fr/ceos:4.33.1F	running	192.168.121.101 N/A
clab-tpgnmic-ceos2	ceos registry.iutbeziers.fr/ceos:4.33.1F	running	192.168.121.102 N/A
clab-tpgnmic-linux1	linux registry.iutbeziers.fr/network-multitool:mai2025	running	192.168.121.103 N/A
clab-tpgnmic-linux2	linux registry.iutbeziers.fr/network-multitool:mai2025	running	192.168.121.104 N/A

```
(netlab) test@0-179:~/gnmic/tpgnmic$ netlab connect ceos1
Connecting to clab-tpgnmic-ceos1 using SSH port 22
Last login: Tue Jun 10 12:49:17 2025 from 192.168.121.1
ceos1#
```

4 Utilisation de gnmic

4.1 Configuration de gnmic

```
(netlab) test@0-179:~/gnmic/tpgnmic$ gnmic -a 192.168.121.102:6030 -u admin -p admin --insecure capabilities
gNMI version: 0.7.0
supported models:
- arista-system-session, Arista Networks <http://arista.com/>, 0.1.0
- arista-bgp-deviations, Arista Networks, Inc., 1.28.0
- arista-sand-control-plane-traffic-counters, Arista Networks <http://arista.com/>, 1.0.0
- openconfig-rib-bgp, OpenConfig working group, 0.9.0
- openconfig-platform-software, OpenConfig working group, 0.1.1
- gnsi-credentialz, Google LLC,
- arista-sand-hardware-drop-counters, Arista Networks <http://arista.com/>, 1.0.0
- openconfig-segment-routing-types, OpenConfig working group, 0.2.0
- arista-arnet-vrf-name, Arista Networks <http://arista.com/>, 0.1.0
- openconfig-vlan, OpenConfig working group, 3.2.2
- arista-ale-counters-open-config-specialized-counters, Arista Networks <http://arista.com/>, 1.0.0
- openconfig-evpn-types, OpenConfig working group, 0.2.0
- openconfig-macsec-types, OpenConfig working group, 0.1.0
- openconfig-aft-types, OpenConfig Working Group, 1.1.0
- openconfig-srte-policy, OpenConfig working group, 0.2.3
- openconfig-programming-errors, OpenConfig working group, 0.1.0
- openconfig-platform-types, OpenConfig working group, 1.8.0
- arista-vlan-augments, Arista Networks <http://arista.com/>, 1.1.2
- openconfig-telemetry, OpenConfig working group, 0.5.1
- openconfig-platform, OpenConfig working group, 0.30.0
```

5 Utilisation de gnmic

1. Créez un fichier de configuration gnmic.yaml dans la directory projet afin d'éviter de resaisir les paramètres de connexion.

```
GNU nano 7.2
targets:
  ceos1:
    address: 192.168.121.101:6030
    username: admin
    password: admin
    insecure: true
```

Vous travaillerez sur routeur ceos1 pur les questions suivantes.

1. Récupérer les capacités du routeur ceos1.

```
(netlab) test@0-179:~/gnmic/tpgnmic$ gnmic --config gnmic.yaml capabilities
gNMI version: 0.7.0
supported models:
- openconfig-sampling, OpenConfig working group, 0.1.0
- openconfig-yang-types, OpenConfig working group, 1.0.0
- openconfig-alarm-types, OpenConfig working group, 0.2.1
- openconfig-platform-types, OpenConfig working group, 1.8.0
- openconfig-bgp, OpenConfig working group, 9.8.0
- arista-sflow-augments, Arista Networks, Inc., 1.1.0
- openconfig-interfaces, OpenConfig working group, 3.7.1
- openconfig-hercules-platform, OpenConfig Hercules Working Group, 0.2.0
- openconfig-metadata, OpenConfig working group, 0.1.0
```

2. Récupérer l'état de l'interface Ethernet1

```
(netlab) test@0-179:~/gnmic/tpgnmic$ gnmic --config gnmic.yaml get --path /interfaces/interface[name=Ethernet1]/state
[
  {
    "source": "ceos1",
    "timestamp": 1749560265673444865,
    "time": "2025-06-10T14:57:45.673444865+02:00",
    "updates": [
      {
        "Path": "interfaces/interface[name=Ethernet1]/state",
        "values": {
          "interfaces/interface/state": {
            "arista-intf-augments:inactive": false,
            "openconfig-interfaces:admin-status": "UP",
            "openconfig-interfaces:counters": {
              "carrier-transitions": "4",
              "in-broadcast-pkts": "5",
              "in-discards": "0",
              "in-errors": "1",
              "in-fcs-errors": "0",
              "in-multicast-pkts": "22",
              "in-octets": "7965",
              "in-pkts": "54",
              "in-unicast-pkts": "27",
              "out-broadcast-pkts": "0",
              "out-discards": "0",
              "out-errors": "0",
              "out-multicast-pkts": "0",
              "out-octets": "0",
              "out-pkts": "0",
              "out-unicast-pkts": "0"
            }
          }
        }
      }
    ]
  }
]
```

3. Récupérer les compteurs de paquets de l'interface Ethernet1

```
(netlab) test@0-179:~/gnmic/tpgnmic$ gnmic --config gnmic.yaml get --path /interfaces/interface[name=Ethernet1]/state/counters
[
  {
    "source": "ceos1",
    "timestamp": 1749560291443169744,
    "time": "2025-06-10T14:58:11.443169744+02:00",
    "updates": [
      {
        "Path": "interfaces/interface[name=Ethernet1]/state/counters",
        "values": {
          "interfaces/interface/state/counters": {
            "openconfig-interfaces:carrier-transitions": "4",
            "openconfig-interfaces:in-broadcast-pkts": "5",
            "openconfig-interfaces:in-discards": "0",
            "openconfig-interfaces:in-errors": "1",
            "openconfig-interfaces:in-fcs-errors": "0",
            "openconfig-interfaces:in-multicast-pkts": "23",
            "openconfig-interfaces:in-octets": "8389",
            "openconfig-interfaces:in-pkts": "57",
            "openconfig-interfaces:in-unicast-pkts": "29",
            "openconfig-interfaces:out-broadcast-pkts": "0",
            "openconfig-interfaces:out-discards": "0",
            "openconfig-interfaces:out-errors": "0",
            "openconfig-interfaces:out-multicast-pkts": "0",
            "openconfig-interfaces:out-octets": "0",
            "openconfig-interfaces:out-pkts": "0",
            "openconfig-interfaces:out-unicast-pkts": "0"
          }
        }
      }
    ]
  }
]
```

4. Récupérez le "neighbor" BGP

```
(netlab) test00-179:~/gnmic/tpgnmic$ gnmic --config gnmic.yaml get --path /network-instances/network-instance[name=default]/protocols/protocol[name=BGP]/bgp/neighbors
{
  "source": "ceos1",
  "timestamp": 1749560324593480914,
  "time": "2025-06-10T14:58:44.593480914+02:00",
  "updates": [
    {
      "Path": "network-instances/network-instance[name=default]/protocols/protocol[identifier=BGP][name=BGP]/bgp/neighbors",
      "values": {
        "network-instances/network-instance/protocols/protocol/bgp/neighbors": {
          "openconfig-network-instance:neighbor": [
            {
              "afi-safis": {
                "afi-safi": [
                  {
                    "afi-safi-name": "arista-bgp-augments:IPv4_MULTICAST",
                    "config": {
                      "afi-safi-name": "arista-bgp-augments:IPv4_MULTICAST"
                    },
                    "state": {
                      "active": false,
                      "afi-safi-name": "arista-bgp-augments:IPv4_MULTICAST"
                    }
                  },
                  {
                    "afi-safi-name": "arista-bgp-augments:IPv6_MULTICAST",
                    "config": {
                      "afi-safi-name": "arista-bgp-augments:IPv6_MULTICAST"
                    },
                    "state": {
                      "active": false,
                      "afi-safi-name": "arista-bgp-augments:IPv6_MULTICAST"
                    }
                  }
                ]
              },
              {
                "afi-safi-name": "openconfig-bgp-types:IPv4_UNICAST",
                "config": {
                  "afi-safi-name": "openconfig-bgp-types:IPv4_UNICAST",
                  "enabled": true
                }
              }
            ]
          }
        }
      }
    }
  ]
}
```

5. Récupérez l'état de la session BGP

```
(netlab) test00-179:~/gnmic/tpgnmic$ gnmic --config gnmic.yaml get --path /network-instances/network-instance[name=default]/protocols/protocol[name=BGP]/bgp/neighbors
{
  "source": "ceos1",
  "timestamp": 1749560431099361907,
  "time": "2025-06-10T15:00:31.099361907+02:00",
  "updates": [
    {
      "Path": "network-instances/network-instance[name=default]/protocols/protocol[identifier=BGP][name=BGP]/bgp/neighbors",
      "values": {
        "network-instances/network-instance/protocols/protocol/bgp/neighbors": {
          "openconfig-network-instance:neighbor": [
            {
              "afi-safis": {
                "afi-safi": [
                  {
                    "afi-safi-name": "arista-bgp-augments:IPv4_MULTICAST",
                    "config": {
                      "afi-safi-name": "arista-bgp-augments:IPv4_MULTICAST"
                    },
                    "state": {
                      "active": false,
                      "afi-safi-name": "arista-bgp-augments:IPv4_MULTICAST"
                    }
                  },
                  {
                    "afi-safi-name": "arista-bgp-augments:IPv6_MULTICAST",
                    "config": {
                      "afi-safi-name": "arista-bgp-augments:IPv6_MULTICAST"
                    },
                    "state": {
                      "active": false,
                      "afi-safi-name": "arista-bgp-augments:IPv6_MULTICAST"
                    }
                  }
                ]
              },
              {
                "afi-safi-name": "openconfig-bgp-types:IPv4_UNICAST",
                "config": {
                  "afi-safi-name": "openconfig-bgp-types:IPv4_UNICAST",
                  "enabled": true
                }
              }
            ]
          }
        }
      }
    }
  ]
}
```

6. Récupérez toutes les informations de l'équipement.

gnmic --config gnmic.yaml get --path /

```

    "mount-point": [
      {
        "name": "/",
        "state": {
          "utilized": "164859376"
        }
      }
    ]
  },
},
{
  "Path": "",
  "values": {
    "": {
      "openconfig-system:system": {
        "mount-points": {
          "mount-point": [
            {
              "name": "/var/lib/docker"
            }
          ]
        }
      }
    }
  }
},
{
  "Path": "",
  "values": {
    "": {
      "openconfig-system:system": {
        "mount-points": {
          "mount-point": [
            {
              "name": "/var/lib/docker",
              "state": {
                "name": "/var/lib/docker"
              }
            }
          ]
        }
      }
    }
  }
}

```

7. Récupérez l'état de la mémoire de l'équipement.

```
(netlab) test@0-179: /gnmic/cpgnmi-$ gnmic --config gnmic.yaml get --path /system/memory
{
  "source": "ceos1",
  "timestamp": 1749560735466106832,
  "time": "2025-06-10T15:05:35.466106832+02:00",
  "updates": [
    {
      "Path": "system/memory",
      "values": {
        "system/memory": {
          "arista-system-augments:kernel": {
            "state": {
              "active": "6953050112",
              "anon-pages": "11943829504",
              "bounce-buffers": "0",
              "buffers": "1058623488",
              "cached": "16153112576",
              "commit-limit": "17713008640",
              "committed-as": "1180999430144",
              "direct-map-1g": "7516192768",
              "direct-map-2m": "26895974400",
              "direct-map-4k": "746008576",
              "dirty": "1945600",
              "inactive": "22336925696",
              "kernel-stack": "47775744",
              "mapped": "7504117760",
              "page-tables": "183500800",
              "shmem": "2638401536",
              "slab": "1861935104",
              "slab-reclaimable": "1434345472",
              "slab-unreclaim": "427589632",
              "swap-cached": "28192768",
              "swap-free": "377536512",
              "swap-total": "1023406080",
              "vmalloc-total": "35184372087808",
              "writeback": "0",
              "writeback-tmp": "0"
            }
          }
        }
      },
    }
  ]
}
```

8. Récupérez le nombre de CPU de l'équipement au format "flat"

```
(netlab) test00-179:~/gnmic/tpgmic$ gnmic --config gnmic.yaml get --path /system/cpus/cpu
[
  {
    "source": "ceos1",
    "timestamp": 1749561546125400044,
    "time": "2025-06-10T15:19:06.125400044+02:00",
    "updates": [
      {
        "Path": "system/cpus/cpu[index=0]",
        "values": {
          "system/cpus/cpu": {
            "openconfig-system:index": 0,
            "openconfig-system:state": {
              "hardware-interrupt": {
                "avg": 0,
                "instant": 0,
                "interval": "1000000000000",
                "max": 0,
                "max-time": "1749561540856309175",
                "min": 0,
                "min-time": "1749561540856309175"
              },
              "idle": {
                "avg": 95,
                "instant": 96,
                "interval": "1000000000000",
                "max": 98,
                "max-time": "1749561410856338739",
                "min": 90,
                "min-time": "1749561020856791257"
              }
            }
          },
          "index": 0,
          "kernel": {
            "avg": 1,
            "instant": 1,
            "interval": "1000000000000",
            "max": 4,
            "max-time": "1749561530856082201",
            "min": 0,
            "min-time": "1749561450855746746"
          }
        }
      }
    ]
  }
]
```

9. Explorer l'API avec gnmic prompt

```
(netlab) test00-179:~/gnmic/tpgmic$ gnmic prompt --config gnmic.yaml
gnmic>
capabilities      query targets gnmic capabilities
completion        generate completion script
diff              run a diff comparison between targets
generate          generate paths or JSON/YAML objects from YANG
get               run gnmic get on targets
getset            run gnmic get then set on targets
help              Help about any command
listen            listens for telemetry dialout updates from the node
output            manipulate configured outputs
path              generate gnmic or xpath style from yang file
```

10. Souscrivez à l'état des compteurs de l'interface Ethernet1 en échantillonnant toutes les 5 secondes.

```
python3 --content: /home/test-tpgmic/telemetry/telemetry.py
(netlab) test00-179:~/gnmic/tpgmic$ gnmic --config gnmic.yaml subscribe --path /interfaces/interface[name=Ethernet1]/state/counters --stream-mode sample --sample-interval 5s
[
  {
    "source": "ceos1",
    "subscription-name": "default-1749562046",
    "timestamp": 174959741602659986,
    "time": "2025-06-10T14:49:23.135713331+02:00",
    "updates": [
      {
        "Path": "interfaces/interface[name=Ethernet1]/state/counters/in-broadcast-pkts",
        "values": {
          "interfaces/interface/state/counters/in-broadcast-pkts": 5
        }
      }
    ]
  },
  {
    "source": "ceos1",
    "subscription-name": "default-1749562046",
    "timestamp": 174959741602659986,
    "time": "2025-06-10T14:49:01.602659986+02:00",
    "prefix": "interfaces/interface[name=Ethernet1]/state/counters",
    "updates": [
      {
        "Path": "in-discards",
        "values": {
          "in-discards": 0
        }
      },
      {
        "Path": "in-fcs-errors",
        "values": {
          "in-fcs-errors": 0
        }
      }
    ]
  }
]
```



```

"source": "ceos1",
"subscription-name": "default-1749562046",
"timestamp": 1749562051695549852,
"time": "2025-06-10T15:27:31.695549852+02:00",
"updates": [
  {
    "Path": "interfaces/interface[name=Ethernet1]/state/counters/in-unicast-pkts",
    "values": {
      "interfaces/interface/state/counters/in-unicast-pkts": 98
    }
  }
]
}
{
"source": "ceos1",
"subscription-name": "default-1749562046",
"timestamp": 174959750616738822,
"time": "2025-06-10T14:49:10.616738822+02:00",
"updates": [
  {
    "Path": "interfaces/interface[name=Ethernet1]/state/counters/in-errors",
    "values": {
      "interfaces/interface/state/counters/in-errors": 1
    }
  }
]
}
}
{
"source": "ceos1",
"subscription-name": "default-1749562046",
"timestamp": 1749562098769993210,
"time": "2025-06-10T15:28:10.76999321+02:00"
}

```

11. Récupérer les routes BGP

```

(netlab test@0-179:~/gnmic/tpgnmic$ gnmic --config gnmic.yaml get --path /network-instances/network-instance[name=default]/protocols/protocol[name=BGP]/bgp/global/arista-bgp-augments:networks
[
  {
    "source": "ceos1",
    "timestamp": 1749562050716448414,
    "time": "2025-06-10T15:40:50.716448414+02:00",
    "updates": [
      {
        "Path": "network-instances/network-instance[name=default]/protocols/protocol[identifier=BGP][name=BGP]/bgp/global/networks",
        "values": {
          "network-instances/network-instance/protocols/protocol/bgp/global/networks": {
            "arista-bgp-augments:network": [
              {
                "config": {
                  "network": "10.0.0.1/32"
                },
                "network": "10.0.0.1/32",
                "state": {
                  "network": "10.0.0.1/32"
                }
              },
              {
                "config": {
                  "network": "172.16.0.0/24"
                },
                "network": "172.16.0.0/24",
                "state": {
                  "network": "172.16.0.0/24"
                }
              }
            ]
          }
        }
      }
    ]
  }
]

```

12. Récupérer les processus de l'équipement. Quel est leur nombre?

```

(netlab test@0-179:~/gnmic/tpgnmic$ gnmic --config gnmic.yaml get --path /system/processes --format flat
system/processes/openconfig-system:process.0/pid: 1
system/processes/openconfig-system:process.0/state/args.0: systemd.setenv=SKIP_ZERO_TOUCH_BARRIER_IN_SYSDDBINIT=1 systemd.setenv=INTFTYPE=et systemd.setenv=MGMT_INTF=eth0 systemd.setenv=no_proxy=localhost,127.0.0.1,::1,* local,192.168.121.0/24,192.168.121.101,192.168.121.102,192.168.121.103,192.168.121.104,ceos1,ceos2,linux1,linux2 systemd.setenv=CEOS=1 systemd.setenv=EOS_PLATFORM=ceoslab systemd.setenv=container=docker systemd.setenv=ETDA=1 systemd.setenv=CLAB_MGMT_VRF=management systemd.setenv=MAPETHE=1 systemd.setenv=NO_PROXY=localhost,127.0.0.1,::1,* local,192.168.121.0/24,192.168.121.101,192.168.121.102,192.168.121.103,192.168.121.104,ceos1,ceos2,linux1,linux2
system/processes/openconfig-system:process.0/state/cpu-usage-system: 19
system/processes/openconfig-system:process.0/state/cpu-usage-user: 24
system/processes/openconfig-system:process.0/state/cpu-utilization: 0
system/processes/openconfig-system:process.0/state/memory-usage: 14180352
system/processes/openconfig-system:process.0/state/memory-utilization: 0
system/processes/openconfig-system:process.0/state/name: systemd
system/processes/openconfig-system:process.0/state/pid: 1
system/processes/openconfig-system:process.0/state/start-time: 1750858847593048095
system/processes/openconfig-system:process.1/pid: 79
system/processes/openconfig-system:process.1/state/args.0:
system/processes/openconfig-system:process.1/state/cpu-usage-system: 5
system/processes/openconfig-system:process.1/state/cpu-usage-user: 9
system/processes/openconfig-system:process.1/state/cpu-utilization: 0
system/processes/openconfig-system:process.1/state/memory-usage: 10424320
system/processes/openconfig-system:process.1/state/memory-utilization: 0
system/processes/openconfig-system:process.1/state/name: systemd-journal
system/processes/openconfig-system:process.1/state/pid: 79
system/processes/openconfig-system:process.1/state/start-time: 1750858848593048095
system/processes/openconfig-system:process.10/pid: 423
system/processes/openconfig-system:process.10/state/args.0: /usr/bin/lnotifyrun -c date; pax -x sv4cpio -O -w -f /mnt/flash/persist/local.new . && mv /mnt/flash/persist/local.new /mnt/flash/persist/local || log
ex -t SetupPersist Failed to update /mnt/flash/persist/local --daemon --ratelimit 0.5 --logfile=/var/log/lnotifyrun-local.log --pidfile=/var/run/lnotifyrun-local.pid /persist/local
system/processes/openconfig-system:process.10/state/cpu-usage-system: 0

```

13. Récupérer la description de l'interface Ethernet1, changez la en "effaçable" puis supprimez-la.

```
(netlab) test00-179: /gnmic/tpgnmic$ # Changer la description
gnmic --config gnmic.yaml set --update-path /interfaces/interface[name=Ethernet1]/config/description --update-value "effaçable"

# Supprimer la description
gnmic --config gnmic.yaml set --delete /interfaces/interface[name=Ethernet1]/config/description
{
  "source": "ceos1",
  "timestamp": 1749563863982619240,
  "time": "2025-06-10T15:44:23.98261924+02:00",
  "results": [
    {
      "operation": "UPDATE",
      "path": "interfaces/interface[name=Ethernet1]/config/description"
    }
  ]
}
{
  "source": "ceos1",
  "timestamp": 1749563864476252886,
  "time": "2025-06-10T15:44:24.476252886+02:00",
  "results": [
    {
      "operation": "DELETE",
      "path": "interfaces/interface[name=Ethernet1]/config/description"
    }
  ]
}
(netlab) test00-179: /gnmic/tpgnmic$
```

14. utiliser l'événement on_change pour être prévenu de la suppression de l'AS BGP.

```
(netlab) test00-179: /gnmic/tpgnmic$ gnmic --config gnmic.yaml subscribe --path /network-instances/network-instance[name=default]/protocols/protocol[name=BGP]/bgp/neighbors/neighbor/state/session-state --stream-mode on_change
{
  "source": "ceos1",
  "subscription-name": "default-1749563873",
  "timestamp": 1749563281766746191,
  "time": "2025-06-10T15:48:01.766746191+02:00",
  "updates": [
    {
      "Path": "network-instances/network-instance[name=default]/protocols/protocol[identifier=BGP][name=BGP]/bgp/neighbors/neighbor[neighbor-address=10.1.0.2]/state/session-state",
      "values": {
        "network-instances/network-instance/protocols/protocol/bgp/neighbors/neighbor/state/session-state": "ESTABLISHED"
      }
    }
  ]
}
{
  "sync-response": true
}
{
  "source": "ceos1",
  "subscription-name": "default-1749563873",
  "timestamp": 1749563976891783877,
  "time": "2025-06-10T15:59:36.891783877+02:00",
  "updates": [
    {
      "Path": "network-instances/network-instance[name=default]/protocols/protocol[identifier=BGP][name=BGP]/bgp/neighbors/neighbor[neighbor-address=10.1.0.2]/state/session-state",
      "values": {
        "network-instances/network-instance/protocols/protocol/bgp/neighbors/neighbor/state/session-state": "ACTIVE"
      }
    }
  ]
}
```

```
(netlab) test00-179:~/gnmic/tpgnmic$ ssh admin@192.168.121.101
The authenticity of host '192.168.121.101 (192.168.121.101)' can't be established.
ED25519 key fingerprint is SHA256:HUhYYPjg0i+QBSLRRcgVMnAtCuBvRW7cKiu7De0OHNA.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.121.101' (ED25519) to the list of known hosts.
(admin@192.168.121.101) Password:
Last login: Tue Jun 10 13:49:23 2025 from 192.168.121.1
ceos1#configure
ceos1(config)#router bgp 65100
ceos1(config-router-bgp)#router-id 10.0.0.2
ceos1(config-router-bgp)#commit
% Unavailable command (no active session)
ceos1(config-router-bgp)#router-id 10.0.0.1
ceos1(config-router-bgp)#commit
% Unavailable command (no active session)
ceos1(config-router-bgp)#end
ceos1#wr
Copy completed successfully.
ceos1#configure
ceos1(config)#router bgp 65100
ceos1(config-router-bgp)#router-id 10.0.0.2
ceos1(config-router-bgp)#exit
ceos1(config)#exit
ceos1#write memory
Copy completed successfully.
ceos1#
```

15. Faire un "shut" de l'interface Ethernet1 et un "no shut".

```

(netlab) test@0-179:~/gnmic/tpggnmic$ # Shut
gnmic --config gnmic.yaml set --update-path /interfaces/interface[name=Ethernet1]/config/enabled --update-value false

# No shut
gnmic --config gnmic.yaml set --update-path /interfaces/interface[name=Ethernet1]/config/enabled --update-value true
{
  "source": "ceos1",
  "timestamp": 1749564084674398318,
  "time": "2025-06-10T16:01:24.674398318+02:00",
  "results": [
    {
      "operation": "UPDATE",
      "path": "interfaces/interface[name=Ethernet1]/config/enabled"
    }
  ]
}
{
  "source": "ceos1",
  "timestamp": 1749564085198135484,
  "time": "2025-06-10T16:01:25.198135484+02:00",
  "results": [
    {
      "operation": "UPDATE",
      "path": "interfaces/interface[name=Ethernet1]/config/enabled"
    }
  ]
}

```

6 Gnmi et Prometheus

Paramétrez gnmic.yaml et prometheus.yml afin que gnmic puisse être utilisé comme un exporter

Prometheus

```
test@0-179:~/gnmic/tpgnmic$ cat gnmic.yaml
targets:
  ceos1:
    address: 192.168.121.101:6030
    username: admin
    password: admin
    insecure: true

subscriptions:
  eth1-counters:
    paths:
      - /interfaces/interface[name=Ethernet1]/state/counters
    stream-mode: sample
    sample-interval: 10s

outputs:
  prometheus:
    type: prometheus
    listen: ":9804"

logging:
  level: info
  format: json

api-server:
  enable-metrics: true
  enable-pprof: true
  address: ":9804"
```

```
test@0-179:~/gnmic/tpgnmic$ cat prometheus.yml
global:
  scrape_interval: 10s

scrape_configs:
  - job_name: 'gnmic'
    static_configs:
      - targets: ['localhost:9804']
```

```
(netlab) test@0-179:~/gnmic/tpgnmic$ gnmic --config gnmic.yaml subscribe
```

```
(netlab) test@0-179:~/gnmic/tpgnmic$ docker run --rm --network host \
-v "$(pwd)/prometheus.yml:/etc/prometheus/prometheus.yml" \
prom/prometheus
```

